Explicit Local and Recurrent History Correction for Event Quantity Prediction

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Abstract. An event encoder can summarize earlier observations, but this does not establish whether a separate path from recent observations improves quantity prediction. We examine this question in a two-block causal encoder for next-event quantity and recorded-time prediction. A local three-position correction modifies the first block's query, key, and value projections; a recurrent residual connects the first and second blocks. Controls remove this path or replace its cross-position inputs with current-position operations at the same active parameter count. A separate study tests fixed combination of time- and quantity-conditioned prototype scores against uniform, similarity-weighted, and parameter-matched general-query retrieval. The protocol fixes observable histories, output heads, checkpoint selection, and training rules on three core datasets; Instacart supplies a supplementary architecture comparison. The new representative comparisons and independent evaluation are pending. This draft therefore states hypotheses and protocol, without a superiority claim.

Keywords: Event sequences · Quantity forecasting · History correction · Temporal point processes

1 Introduction

Irregular demand sequences require both next-event quantity and recorded time to be predicted. Recurrent and attention-based encoders already condition these predictions on event history. We ask whether an additional explicit local and recurrent path improves quantity prediction within such an encoder. A full correction is compared with its removal and with a current-position control of equal active parameter count. The latter retains shared causal attention, so it remains history-aware.

A secondary question asks whether separate time- and quantity-query normalization, followed by fixed score combination, improves prototype retrieval. We compare it with top-four averaging, similarity weighting, and a general query under the same information and extra-parameter budget. The prototype bank is learned on Train and stays static during evaluation; it is not an online store of observed events.

A completed MLP8 validation study motivates these questions but does not prove them for the representative CNN+GRU54 architecture. Its matched

history-on/off pairs improved overall quantity RMSE in 18 of 18 cases, yet this comparison also changed capacity and some quantity-selected checkpoints worsened timing likelihood. The new study retains unfavorable quantity-tail, time, and cost outcomes. It uses one representative model specification across the core datasets rather than choosing a different winner for each dataset.

2 Related Work

RMTPP summarizes event history in recurrent hidden vectors [1]; deep renewal models predict demand size and interdemand time from recurrent representations [2]. Attention-based A-NHP models irregular event histories with temporal attention and residual layers [3]. S2P2 combines continuous-time dynamics with event impulses and nonlinear residual transformations [4]. History, residual connections, and joint time–quantity modeling are thus established ideas. An event quantity can itself be regarded as a mark even without a categorical event-type ID.

Persistent-memory attention adds learned, input-independent vectors [5], and product-key memory selects learned slots sparsely [6]. The present study does not claim novelty for top-k retrieval or learned static slots. Its retrieval hypothesis concerns the specified split-query constraint under matched budgets.

For a later external comparison, the inspected S2P2 adapter projects observed numerical gaps and quantities into the S2P2 history encoder, reads its state immediately after the last observed event, and attaches the shared prediction heads. It is a common-head history-encoder adaptation, not a reproduction of S2P2's native marked likelihood. Historical external runs require separate source, population, Runtime, head, and selection audits before they can substantiate a claim about the representative architecture.

3 Prediction Task and Shared Heads

Let $\mathcal{H}_j = \{(\Delta_i, q_i)\}_{i \le j}$ be the observed prefix. Position j is the last observed event, and $j+1$ the withheld target. Its quantity and gap do not enter the TitanTPP encoder readout. The input embedding is

$$e_i = W_{\text{in}}[\log(1+\Delta_i), \log(1+q_i)]^\top + b_{\text{in}} + p_i. \tag{1}$$

Here p_i is learned positional information. These features are not Train-z-scored. Two pre-normalized causal attention blocks have width 64, four heads, feedforward width 128, dropout 0.1, and 16 learned persistent attention tokens per block. The persistent tokens differ from the 64 prototype vectors applied after the encoder.

For final representation z_j, the quantity head produces

$$\widehat{\ell}_{j+1} = \text{softplus}(w_q^\top z_j + b_q), \qquad \widehat{q}_{j+1} = \exp(\widehat{\ell}_{j+1}) - 1. \tag{2}$$

Training compares $\widehat{\ell}_{j+1}$ with $\log(1+q_{j+1})$ by squared error. Raw-unit evaluation uses Eq. 2, with no later inverse Train standardization.

The timing head predicts location μ_j and scale $\sigma_j = 10^{-3} + \mathrm{softplus}(w_\sigma^\top z_j + b_\sigma)$. A latent duration $T = s\exp(\mu_j + \sigma_j\epsilon)$, $\epsilon \sim \mathcal{N}(0,1)$, is recorded as the positive integer $D = \max(1, \mathrm{round}(T))$. For standard normal CDF Φ, its probability mass is

$$p_j(1) = \Phi\left(\frac{\log(1.5/s) - \mu_j}{\sigma_j}\right), \tag{3}$$

$$p_j(d) = \Phi\left(\frac{\log((d+0.5)/s) - \mu_j}{\sigma_j}\right) - \Phi\left(\frac{\log((d-0.5)/s) - \mu_j}{\sigma_j}\right), \quad d \geq 2. \tag{4}$$

The core datasets have no upper top-code. Instacart top-codes at 30 and uses the upper survival mass. The target loss adds $-\log p_j(D_{j+1})$ to the log-quantity squared error, with unit quantity coefficient and no tail-weighted loss.

4 Explicit History Correction

The first block projects normalized inputs into queries, keys, and values. For each $X \in \{Q, K, V\}$, it applies

$$\widetilde{X}_i = X_i + \sum_{\ell=0}^{2} a_{i\ell}\kappa_\ell^X \odot X_{s_\ell(i)}. \tag{5}$$

Here $s_0(i) = i$; $s_1(i)$ and $s_2(i)$ select the preceding two observed positions in the current segment. Eligibility $a_{i\ell}$ requires at least $\ell + 1$ observed positions. Each depthwise kernel has 64 bias-free coefficients. Persistent key and value tokens enter after this local operation. Let b_i be the complete first attention block output, including its feedforward residual.

A width-54 recurrent path connects the blocks:

$$u_i = Ab_i, \quad r_i = \mathrm{GRU}(u_i, r_{i-1}), \quad c_i = \mathbf{1}[n_i \geq 2]Br_i, \quad \widetilde{b}_i = b_i + c_i. \tag{6}$$

The matrices map $64 \to 54$ and $54 \to 64$. The count n_i includes the current observed position in its segment; recurrent state begins at zero at each segment boundary. The second block receives $\widetilde{b}_i$. Thus the residual contains the recurrent state updated by the current observation. Padding contributes no correction; a valid withheld position resets local eligibility and recurrent state. This reset does not introduce an additional segment mask into the shared causal attention.

The full path contains $3 \cdot 3 \cdot 64 = 576$ local coefficients and 24,732 recurrent-path parameters: 25,308 active correction parameters in total. *Disabled* bypasses and freezes the path while preserving both attention blocks, persistent tokens, mean prototype retrieval, and prediction heads. *Current-only* retains the same correction tensors. Its three local source values all come from the current position:

$$X_i, \quad \tanh(X_i), \quad \mathrm{GELU}(X_i).$$

Its recurrent state starts at zero independently at each position and processes the current u_i twice, with no state carried between positions. This allows its recurrent weights to participate. The first-block b_i already contains prefix context, whereas the local X_i is computed before the first attention operation. The equal parameter count does not equalize functional capacity, depth, or cost. A positive comparison would support the combined three-position and recurrent correction path; it would not isolate lag one, CNN, or GRU alone.

5 Static Prototype Retrieval

Let $M = \{m_k\}_{k=1}^{64}$ be train-learned vectors of width 64. Define normalization by

$$\mathrm{N}(v) = \frac{v}{\max(\|v\|_2, \varepsilon)}.$$

The normalized slot supplies its key; the same unnormalized slot supplies its value. For second-block output h_j, mean retrieval scores each slot by

$$s_{jk} = \mathrm{N}(h_j)^\top \mathrm{N}(m_k).$$

It selects the four highest-scoring slots I_j and returns

$$z_j^{\mathrm{mean}} = h_j + \tfrac{1}{4} \sum_{k \in I_j} m_k. \tag{7}$$

Weighted retrieval replaces $\frac{1}{4}$ with softmax weights over those same four scores. Neither arm adds query parameters.

The learned-query arms summarize the most recent 32 observed positions in the current segment, including the current event. Each of time and quantity contributes four features: last standardized value, mean, population standard deviation, and last-minus-previous difference. Positive time gaps use $\log(\Delta_i/s)$; quantities use $\log(1+q_i)$, including zero quantities. Train statistics standardize these points. Missing features are zero. Write the views as $\phi_j^t, \phi_j^q \in \mathbb{R}^4$.

The fixed arm learns $W_t, W_q \in \mathbb{R}^{4\times 64}$ and scores

$$\begin{aligned} q_j^t &= h_j + \phi_j^t W_t, \quad q_j^q = h_j + \phi_j^q W_q, \\ s_{jk}^{\mathrm{fixed}} &= \tfrac{1}{2}\,\mathrm{N}(q_j^t)^\top \mathrm{N}(m_k) + \tfrac{1}{2}\,\mathrm{N}(q_j^q)^\top \mathrm{N}(m_k). \end{aligned} \tag{8}$$

The general arm learns $W \in \mathbb{R}^{8\times 64}$:

$$q_j = h_j + \tfrac{1}{2}[\phi_j^t, \phi_j^q]W, \quad s_{jk}^{\mathrm{general}} = \mathrm{N}(q_j)^\top \mathrm{N}(m_k). \tag{9}$$

Both add exactly 512 active parameters, select four slots, and use a temperature-one softmax for value aggregation. An earlier 521-parameter general-query pilot is not this matched control. The fixed score also equals one-half of the dot product of the concatenated separately normalized queries and duplicated normalized key. This algebraic relation narrows the hypothesis to the effect of the normalization and query constraint; the labels "time" and "quantity" do not establish semantic disentanglement. Every arm is fitted end to end, so fitted encoder and slot values can differ.

Table 1. Frozen task populations and windows. "Large" is strictly above the Train boundary.

Dataset	Train	Validation	Cap	Lookback	Large
Taxi	38,393	8,268	256	168 hourly bins	$q > 3449$
Intermittent	393,824	86,285	256	520 weekly bins	$q > 187$
RAF Spare Parts	25,779	6,690	84	84 monthly bins	$q > 200$
Instacart	1,991,192	503,733	64	52 daily bins	$q > 35$

Table 2. Predeclared comparison. Status is the writing-time record, not a live GPU observation.

Stage	Arms and role	New fits/status
Instacart	Width 8/12/CNN+GRU54; architecture context	9; in progress
Correction	Full/disabled/current-only; primary claim	27; pending
Retrieval	Weighted/fixed/general512; full history reused as mean	27; pending

6 Experimental Design

6.1 Datasets, stages, and training

The core study covers Taxi, Intermittent, and RAF Spare Parts. Instacart is supplementary. Table 1 records frozen eligible target populations. The legacy loader field named `lookback_weeks` counts each dataset's native discrete time bins. Its inclusive window ends at the last observed event; the next target is appended before left-truncation to the sequence cap. The cap includes the target. Large-quantity thresholds were fixed from Train.

The campaign specifies 63 new fits in three sequential stages (Table 2). Instacart compares widths 8 and 12 with CNN+GRU54 across seeds 42, 52, and 62: nine fresh fits. Historical width-16 checkpoints are reference material, not new paired runs. The main correction study trains full, disabled, and current-only variants across three core datasets and three seeds: 27 fresh fits. Retrieval trains weighted, fixed, and general512 across the same datasets and seeds: another 27. It reuses the nine new full-correction fits as mean retrieval, exactly once each. Same-dataset and same-seed arms share a GPU and Runtime. Host timing is reported separately.

All fits use batch size 128, AdamW with learning rate 0.001 and weight decay 0.01, gradient clipping at 1, minimum 40 and maximum 300 epochs, and patience 40. There is no warm start or automatic retry. Checkpoint selection retains the first strictly improving finite minimum of overall raw-quantity RMSE on full Validation; ties keep the earlier epoch. Neither a tail score nor time likelihood changes that choice. A resource timeout is not a completed scientific fit.

Table 3. Reserved Validation comparison. All metrics must come from each fit's quantity-selected checkpoint, with per-seed results in the final version.

Comparison	Overall quantity	Large quantity and time
Full vs. correction controls	pending	pending
Fixed vs. retrieval controls	pending	pending
Instacart architectures	pending	pending

6.2 Metrics, cost, and decisions

At the quantity-selected checkpoint, report quantity RMSE, quantity MAE, and recorded-time negative log-likelihood both on all Validation targets and on the large-quantity subset. All six values refer to one selected epoch. Preserve each seed and report three-seed means, sample standard deviations, adverse cases, selected and final saved epochs, fit time, and measured cost. Recorded-time scores from different units are not pooled.

The correction hypothesis receives developmental support only if full has strictly lower overall raw-quantity RMSE than *both* controls for all three seeds in the same two or more core datasets. The retrieval hypothesis requires fixed to beat mean, weighted, and general512 by the corresponding criterion. It also requires cost eligibility on that dataset's GPU for both lengths 64 and 256. Under batch 128, five warm-up and twenty measured iterations, relative to mean, inference time must rise no more than 20%, training-step time no more than 30%, and peak allocation no more than 20%. Technically successful but over-budget arms remain in the report.

The supplementary Instacart study compares architectures; it does not directly test either correction or retrieval contribution. A result below the strongest comparator would constrain a universal superiority claim, while the predeclared core comparisons would still decide the two mechanism questions.

7 Results and Interpretation

Writing-time status. The MLP8 Validation study and a limited parity check between CNN+GRU54 and mean retrieval are completed background evidence. The representative correction and fixed-retrieval comparisons are pending. Tables 3 and 4 contain *pending* entries so that no new performance result is invented. The final version must fill them from verified original checkpoints and report every seed.

8 Discussion and Limitations

An improvement over correction removal alone would leave active capacity as a plausible explanation. The current-only control narrows this uncertainty, although equal parameters do not equalize the operations or computation. Even a

Table 4. Reserved same-GPU cost audit. Each length requires inference, step, and peak allocation.

Arm	Length 64: inference, step, peak	Length 256: inference, step, peak
Mean	pending	pending
Weighted	pending	pending
Fixed	pending	pending
General512	pending	pending

successful comparison would support the specified local-and-recurrent package, not a lag-one-only or CNN-only effect. Likewise, fixed retrieval has an equivalent concatenated dot-product expression, so performance must be attributed narrowly to the tested constraint and protocol.

Development Validation is not untouched evaluation. Earlier observations informed the fixed-retrieval hypothesis, and historical test partitions have prior research exposure. A new independent assessment needs a verified unexposed period or dataset, a frozen model and selector, and an access-history audit before outcomes are read. The 63-fit campaign does not refit external S2P2. Any final external table must distinguish common-head adaptations from native model likelihoods and disclose weaker comparisons.

Quantity selection may worsen timing likelihood or rare large-quantity errors. We therefore retain the same-epoch six metrics and measured costs without setting a favorable time threshold after the fact. The sign-consistency rules do not by themselves establish statistical significance or universal external superiority.

9 Conclusion

This manuscript draft specifies a primary test of explicit local and recurrent correction within an already history-aware event encoder. A separate matched-budget study tests split-query static-prototype retrieval. The eventual empirical contribution depends on verified three-seed comparisons, timing and cost trade-offs, and later independent evaluation. Pending experimental outcomes cannot yet support a final performance claim.

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